

Chapter 38

LCD and Camera Controller (LCD_CAM)

38.1 Overview

The LCD and Camera controller (LCD_CAM) on the ESP32-P4, consisting of an independent LCD control module and a camera control module, is a versatile component designed to facilitate interfacing with both LCDs and cameras.

The LCD module provides an LCD interface that can connect to RGB, Motorola 6800 (MOTO6800), and Intel 8080 (I8080) compatible LCD devices. The LCD module is responsible for sending video data to the display.

The Camera module provides one parallel camera interface that can connect to external digital video port (DVP) cameras in 8/16-bit modes. It manages communication with the camera to capture images or video frames.

The LCD_CAM controller offers flexibility for projects that involve both displaying information on screens and working with cameras for image or video processing.

38.2 Features

LCD_CAM has the following features:

- Supports the following operation modes:
 - LCD master TX mode
 - Camera slave RX mode
 - Camera master RX mode
- Supports simultaneous connection to an external LCD and a camera
- When interfacing with an external LCD, the following is supported:
 - 8/16/24-bit parallel output modes
 - RGB, MOTO6800, and I8080 LCD formats
 - LCD data retrieved from internal memory or external memory via GDMA
- When interfacing with an external camera (i.e., DVP image sensor), the following is supported:
 - 8/16-bit parallel input modes
 - Camera data stored in internal or external memory via GDMA
- Supports interrupts

38.3 Functional Description

38.3.1 Block Diagram

Figure 38.3-1 shows the structure of LCD_CAM which includes:

- One TX control unit (LCD_Ctrl)
- One RX control unit (Camera_Ctrl)
- One asynchronous TX FIFO (Async TX FIFO) for communicating with external devices, e.g., LCDs
- One asynchronous RX FIFO (Async RX FIFO) for communicating with external devices, e.g., cameras
- Two clock generators (LCD_Clock Generator and CAM_Clock Generator) for generating clocks for each module
- Two format converters (RGB/YUV Converter) for converting video data into various formats

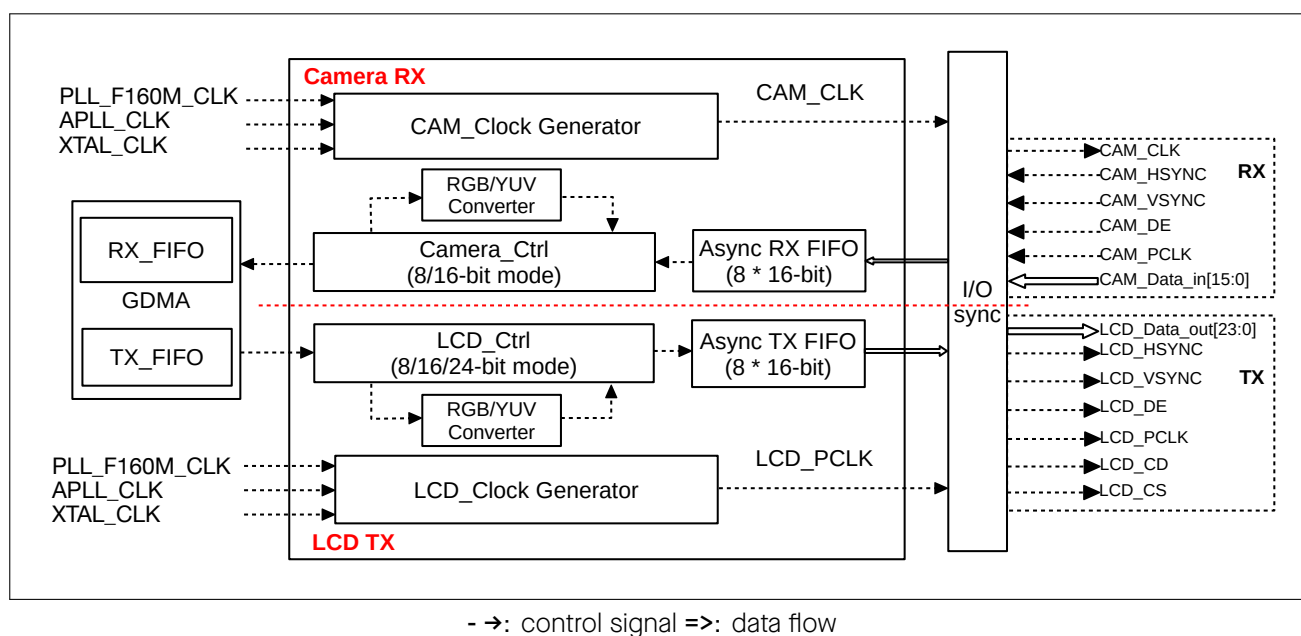


Figure 38.3-1. LCD_CAM Block Diagram

38.3.2 Signal Description

Table 38.3-1 shows the input and output signals required for the operation of the LCD and Camera controller.

The definition of the three operation modes is as follows:

- **Camera Slave RX Mode:** In this mode, the Camera module is the receiver of control signals.
- **Camera Master RX Mode:** In this mode, the Camera module sends clock signals to the slave, which will transmit data to the master upon receiving the signals.
- **LCD Master TX Mode:** In this mode, the LCD module initiates communication with the slave and sends control signals to the slave.

Table 38.3-1. LCD_CAM Input and Output Signal Descriptions

Operation Mode	Signal ¹	Direction	Function
Camera Slave RX Mode	CAM_PCLK	Input	Camera pixel clock signal
	CAM_VSYNC	Input	Vertical synchronization signal (VSYNC) ³
	CAM_HSYNC	Input	Horizontal synchronization signal (HSYNC) ³
	CAM_DE	Input	Horizontal enable signal (DE) ³
	CAM_Data_in[N:0] ²	Input	Camera parallel input data bus, 8/16-bit modes supported
Camera Master RX Mode	CAM_PCLK	Input	Camera pixel clock input signal
	CAM_CLK	Output	Camera master clock output signal
	CAM_VSYNC	Input	Vertical synchronization signal ³
	CAM_HSYNC	Input	Horizontal synchronization signal ³
	CAM_DE	Input	Horizontal enable signal ³
	CAM_Data_in[N:0] ²	Input	Camera parallel input data bus, 8/16-bit modes supported
LCD Master TX Mode	LCD_PCLK	Output	LCD pixel clock signal
	LCD_HSYNC	Output	Horizontal synchronization signal in RGB format
	LCD_VSYNC	Output	Vertical synchronization signal in RGB format
	LCD_DE	Output	Horizontal enable signal in RGB format
	LCD_CD	Output	Command and data (CD) signal in I8080 format
	LCD_CS	Output	Chip select (CS) signal in I8080/MOT06800 format
	LCD_Data_out[M:0] ²	Output	LCD parallel output data bus, 8/16/24-bit modes supported

¹ All the signals must be mapped to the chip's pins via GPIO matrix. For more information, see Chapter 9 [GPIO Matrix and IO MUX](#).

² For the input signals with 8 or 16-bit width, $N = 7$ or 15 respectively. For the output signals with 8, 16, or 24-bit width, $M = 7, 15, \text{ or } 23$ respectively.

³ If [LCD_CAM_CAM_VH_DE_MODE_EN](#) is set, i.e., VSYNC + HSYNC mode is selected, then VSYNC, HSYNC, and DE signals control the data. In this case, users need to wire the three signal lines. If [LCD_CAM_CAM_VH_DE_MODE_EN](#) is cleared, i.e., DE mode is selected, then VSYNC and DE signals control the data. In this case, wiring HSYNC signal line is not a must. But in this case, the YUV-RGB conversion function of the Camera module is not available.

38.3.3 LCD_CAM Module Clocks

38.3.3.1 LCD Clock

The clocks used in the LCD module are generated from clock sources by the LCD_Clock Generator, see Figure 38.3-2. The clocks include:

- Master clock: LCD_CLK, divided from the clock sources
- Pixel clock: LCD_PCLK, divided from LCD_CLK

[HP_SYS_CLKRST_LCD_CLK_EN](#) is used to enable [LCD_CLK_SRC](#) (the clock source), and [HP_SYS_CLKRST_LCD_CLK_SRC_SEL](#) is used to select [LCD_CLK_SRC](#) from one of the following clock sources:

- 0: XTAL_CLK
- 1: PLL_F160M_CLK
- 2: APLL_CLK
- 3: Disable LCD clock source

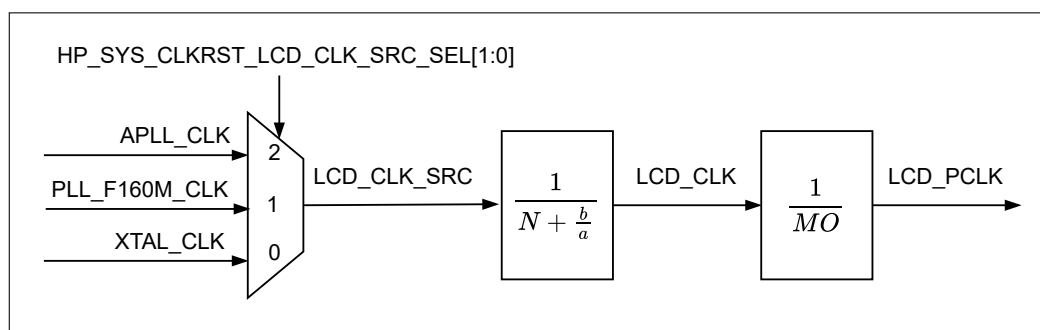


Figure 38.3-2. LCD Clock

The following formula shows the relation between the frequency of [LCD_CLK](#) ($f_{\text{LCD_CLK}}$) and [LCD_CLK_SRC](#) ($f_{\text{LCD_CLK_SRC}}$):

$$f_{\text{LCD_CLK}} = \frac{f_{\text{LCD_CLK_SRC}}}{N + \frac{b}{a}}$$

The values of N , a , and b are related to [HP_SYS_CLKRST_LCD_CLK_DIV_NUM](#), [HP_SYS_CLKRST_LCD_CLK_DIV_NUMERATOR](#), and [HP_SYS_CLKRST_LCD_CLK_DIV_DENOMINATOR](#). Specifically,

- When [HP_SYS_CLKRST_LCD_CLK_DIV_NUM](#) = 0, $N = 1$. The values of a and b do not take effect. The divider is always 1.
- When [HP_SYS_CLKRST_LCD_CLK_DIV_NUM](#) ≥ 1 , $N = \text{HP_SYS_CLKRST_LCD_CLK_DIV_NUM} + 1$.
 - For integer divider, please clear [HP_SYS_CLKRST_LCD_CLK_DIV_NUMERATOR](#) and [HP_SYS_CLKRST_LCD_CLK_DIV_DENOMINATOR](#).
 - For fractional divider, $b = \text{HP_SYS_CLKRST_LCD_CLK_DIV_NUMERATOR}$, $a = \text{HP_SYS_CLKRST_LCD_CLK_DIV_DENOMINATOR}$. The value of [HP_SYS_CLKRST_LCD_CLK_DIV_NUMERATOR](#) should be smaller than the value of [HP_SYS_CLKRST_LCD_CLK_DIV_DENOMINATOR](#).

The following formula shows the relation between the frequencies of [LCD_PCLK](#) ($f_{\text{LCD_PCLK}}$) and [LCD_CLK](#) ($f_{\text{LCD_CLK}}$):

$$f_{\text{LCD_PCLK}} = \frac{f_{\text{LCD_CLK}}}{\text{MO}}$$

MO is determined by [LCD_CAM_LCD_CLK_EQU_SYSCLK](#) and [LCD_CAM_LCD_CLKCNT_N](#), specifically:

- When [LCD_CAM_LCD_CLK_EQU_SYSCLK](#) = 1, MO = 1.

- When `LCD_CAM_LCD_CLK_EQU_SYSCLK = 0`, $MO = LCD_CAM_LCD_CLKCNT_N + 1$.

Notes:

- `LCD_CAM_LCD_CLKCNT_N` must not be configured as 0.
- Using fractional divider may introduce clock jitters. In case `LCD_CLK` and `LCD_PCLK` can not be generated from `PLL_F160M_CLK` by integer divider, `APLL_CLK` can be used as the clock source. For more information, please refer to Chapter 10 [Reset and Clock](#).

38.3.3.2 Camera Clock

The clocks used in the Camera module are generated from clock sources by the CAM_Clock Generator, see Figure 38.3-3. The clocks include:

- Master clock: `CAM_CLK`, master clock output from the Camera module, divided from the clock sources.
- Pixel clock: `CAM_PCLK`, clock input from camera slave.

`HP_SYS_CLKRST_CAM_CLK_EN` is used to enable the `CAM_CLK_SRC` (the clock source), and `HP_SYS_CLKRST_CAM_CLK_SRC_SEL` is used to select `CAM_CLK_SRC` from one of the following clock sources:

- 0: `XTAL_CLK`
- 1: `PLL_F160M_CLK`
- 2: `APLL_CLK`
- 3: Disable camera clock source

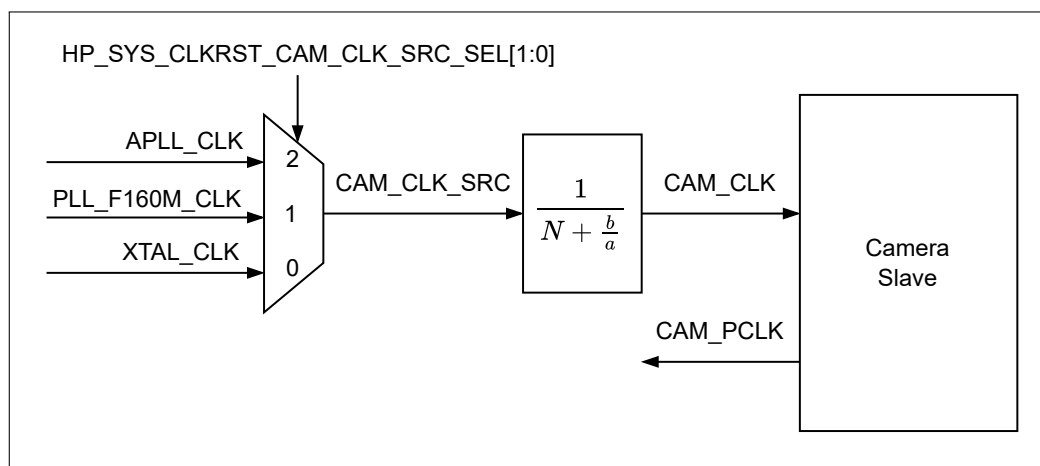


Figure 38.3-3. Camera Clock

The following formula shows the relation between the frequencies of `CAM_CLK` (f_{CAM_CLK}) and the divider's clock source ($f_{CAM_CLK_SRC}$):

$$f_{CAM_CLK} = \frac{f_{CAM_CLK_SRC}}{N + \frac{b}{a}}$$

N is an integer value between 2 and 256. The value of N corresponds to the value of `HP_SYS_CLKRST_CAM_CLK_DIV_NUM` as follows:

- When `HP_SYS_CLKRST_CAM_CLK_DIV_NUM = 0`, $N = 256$.

- When `HP_SYS_CLKRST_CAM_CLK_DIV_NUM` = 1, $N = 2$.
- When `HP_SYS_CLKRST_CAM_CLK_DIV_NUM` = other value, $N = \text{HP_SYS_CLKRST_CAM_CLK_DIV_NUM}$.

b corresponds to the value of `HP_SYS_CLKRST_CAM_CLK_DIV_NUMERATOR`, and a to the value of `HP_SYS_CLKRST_CAM_CLK_DIV_DENOMINATOR`.

- For integer divider, please clear `HP_SYS_CLKRST_CAM_CLK_DIV_NUMERATOR` and `HP_SYS_CLKRST_CAM_CLK_DIV_DENOMINATOR`.
- For fractional divider, the value of `HP_SYS_CLKRST_CAM_CLK_DIV_NUMERATOR` should be smaller than the value of `HP_SYS_CLKRST_CAM_CLK_DIV_DENOMINATOR`.

38.3.4 LCD_CAM Reset

LCD_CAM module has the following reset registers that can reset different parts of the module:

- `HP_SYS_CLKRST_RST_EN_LCDCAM`: Setting this register will reset the whole LCD_CAM module.
- `LCD_CAM_LCD_RESET`: Setting this register will reset the whole LCD module, including the LCD control unit (LCD_Ctrl), the RGB/YUV Converter, and Async TX FIFO. Configuring this register will not affect the Camera module.
- `LCD_CAM_CAM_RESET`: Setting this register will reset the whole Camera module, including the Camera control unit (Camera_Ctrl), the RGB/YUV Converter, and Async RX FIFO. Configuring this register will not affect the LCD module.
- `LCD_CAM_LCD_AFIFO_RESET`: Setting this register will reset the Async TX FIFO without affecting other parts in the module. For example, setting this reset register every time the LCD finishes sending a frame can prevent transmission errors of the previous frame from affecting the transmission of the next frame.
- `LCD_CAM_CAM_AFIFO_RESET`: Setting this register will reset the Async RX FIFO without affecting other parts in the module. For example, setting this reset register every time the Camera finishes receiving a frame can prevent transmission errors of the previous frame from affecting the receiving of the next frame.

Notes:

- The reset register `HP_SYS_CLKRST_RST_EN_LCDCAM` is not self-clearing, meaning writing 1 enables the reset, and writing 0 releases the reset.
- All the above reset registers, except for `HP_SYS_CLKRST_RST_EN_LCDCAM`, are self-clearing, meaning that after writing 1, the hardware will generate a reset pulse and automatically clear the register.
- The module clocks `LCD_CLK` and `CAM_CLK` must be configured first before the module and FIFO are reset.

38.3.5 LCD_CAM Data Format Control

38.3.5.1 LCD Data Format Control

When the data is transmitted by the LCD module, it undergoes three stages of processing:

1. Stage One: Data preprocessing
The LCD module processes the input data from GDMA and sends it to the YUV-RGB converter.

2. Stage Two: Data color space conversion

The YUV-RGB converter processes the data. If the YUV-RGB converter is disabled, the data remains unchanged at this stage.

3. Stage Three: Data post-processing

The LCD module gets data from the YUV-RGB converter for post-processing and then transmits the data to GPIO.

Note:

The output data from the previous stage is the input data for the subsequent stage.

The detailed configuration of each stage is described below.

Stage One: Data Preprocessing

At this stage, configure the following registers to adjust the bit width, bit order, and byte order of the data from GDMA.

- [LCD_CAM_LCD_BYTE_MODE](#)
 - 0: The data width of the LCD input from GDMA is 8 bits.
 - 1: The data width of the LCD input from GDMA is 16 bits.
 - 2: The data width of the LCD input from GDMA is 24 bits.
- [LCD_CAM_LCD_BIT_ORDER](#)
 - 0: Do not invert.
 - 1: Invert the input data bit order.
- [LCD_CAM_LCD_BYTE_ORDER](#)
 - 0: Do not invert.
 - 1: Invert data byte order, only valid in 16/24-bit modes.

For the detailed configuration, see Table [38.3-2](#).

Table 38.3-2. LCD Data Format Control – Stage One

GDMA Data Order ¹	LCD_CAM_LCD_BYTE_MODE	LCD_CAM_LCD_BIT_ORDER	LCD_CAM_LCD_BYTE_ORDER ²	Output Data Order ^{3,4}
B0,B1,B2,B3,B4,B5	0	0	0	{B0}{B1}{B2}{B3}{B4}{B5}
		1	0	{B0'}{B1'}{B2'}{B3'}{B4'}{B5'}
	1	0	0	{B1,B0}{B3,B2}{B5,B4}
			1	{B0,B1}{B2,B3}{B4,B5}
		1	0	{B1',B0'}{B3',B2'}{B5',B4'}
			1	{B0',B1'}{B2',B3'}{B4',B5'}
	2	0	0	{B2,B1,B0}{B5,B4,B3}
			1	{B0,B1,B2}{B3,B4,B5}
		1	0	{B2',B1',B0'}{B5',B4',B3'}
			1	{B0',B1',B2'}{B3',B4',B5'}

¹ B0 ~ B5 represent the bytes of the data from GDMA, from low address to high address.

² Only the configuration listed in the table is valid. Other configurations may cause unexpected data errors.

³ In output data, the bits in {} are in big-endian, and are in parallel with each other, while the data of {} is in serial with the data of other {}. Data is sent out from left to right. Take {B0}{B1}{B2}{B3} as an example. The bits in {B0} are in parallel with each other, but are in serial with the bits in {B1}{B2}{B3}. {B0} is sent out first.

⁴ In output data, Bn'[7:0] = Bn[0:7] (n = 0,1,2,3,4,5).

Stage Two: Data Color Space Conversion

At this stage, the data undergoes color space conversion. For detailed configurations, please refer to Section 38.3.6.

Note:

Users can skip Stage Two by configuring [LCD_CAM_LCD_CONV_ENABLE](#) as 0, in which case the output of Stage One will directly become the input of Stage Three.

Stage Three: Data Post-processing

At this stage, configure the following registers to adjust the bit and byte order of the output data.

- [LCD_CAM_LCD_WIRE_MODE](#)
 - 0: The bit width of the data transmitted to GPIO is 8 bits.
 - 1: The bit width of the data transmitted to GPIO 16 bits.
 - 2: The bit width of the data transmitted to GPIO is 24 bits.
- [LCD_CAM_LCD_DOUT_BIT_ORDER](#)
 - 0: Do not invert.
 - 1: Invert the output data bit order.
- [LCD_CAM_LCD_DOUT_BYTE_SWIZZLE_ENABLE](#)
 - 0: Disable LCD output data byte reordering.
 - 1: Enable LCD output data byte reordering by configuring [LCD_CAM_LCD_DOUT_BYTE_SWIZZLE_MODE](#). For the reordering results see Table 38.3-3.

Table 38.3-3. LCD Output Data Reordering

Stage Three Input Data Byte Order ¹	LCD_CAM_LCD_DOUT_BYTE_SWIZZLE_MODE ²	Stage Three Output Data Byte Order ³
B0,B1,B2,B3,B4,B5	0	B1,B0,B3,B2,B5,B4
	1	B0,B2,B1,B3,B5,B4
	2	B1,B0,B2,B4,B3,B5
	3	B1,B2,B0,B4,B5,B3
	4	B2,B0,B1,B5,B3,B4
	5	B2,B1,B0,B5,B4,B3

¹ B0 ~ B5 represent the bytes of the input data in stage three, i.e., the input data is arranged in a little-endian byte sequence. For example, when the valid data width of the input data is 16 bits, the original input data sequence B1,B0,B3,B2,B5,B4 is converted to B0,B1,B2,B3,B4,B5.

² Only the values listed in the table are valid. Other values may cause unexpected data errors.

³ The byte order of data that is ultimately transmitted to GPIO is determined by the “Stage Three Output Data Byte Order” and the bit width. For example, if the “Stage Three Output Data Byte Order” is B1,B0,B2,B4,B3,B5 and the transmitted data width is 16 bits, then the transmitted data byte order would be {B0,B1}{B4,B2}{B5,B3}. In this sequence, the data within each {} is in big-endian parallel format, and the data in each {} is in serial with the data in other {}. Data is transmitted from left to right. In this particular example, the data in {B0,B1} are in parallel with each other, while the data in {B0,B1} is in serial with the data in {B4,B2} and {B5,B3}. {B0,B1} is transmitted first.

38.3.5.2 Camera Data Format Control

When the Camera module receives data, configure the following registers to adjust the bit/byte order of the data sent to GDMA.

- [LCD_CAM_CAM_2BYTE_EN](#)
 - 0: The data width of the Camera input is 8 bits.
 - 1: The data width of the Camera input is 16 bits.
- [LCD_CAM_CAM_BIT_ORDER](#)
 - 0: Do not invert.
 - 1: Invert data bit order.
 - * Invert CAM_DATA_in[7:0] to CAM_DATA_in[0:7] in 8-bit mode.
 - * Invert CAM_DATA_in[15:0] to CAM_DATA_in[0:15] in 16-bit mode.
- [LCD_CAM_CAM_BYTE_ORDER](#)
 - 0: Do not invert.
 - 1: Invert data byte order, only valid in 16-bit mode.

For the detailed configuration, see Table [38.3-4](#).

Table 38.3-4. Camera Data Format Control

RX Data Order ¹	LCD_CAM_CAM _2BYTE_EN	LCD_CAM_CAM _BIT_ORDER ²	LCD_CAM_CAM _BYTE_ORDER ²	GDMA Order ^{3,4}	Data
{B0}{B1}{B2}{B3}	0	0	0	B0,B1,B2,B3	
		1	0	B0',B1',B2',B3'	
{B1,B0}{B3,B2}	1	0	0	B0,B1,B2,B3	
			1	B1,B0,B3,B2	
		1	0	B0',B1',B2',B3'	
			1	B1',B0',B3',B2'	

¹ In RX data, the bits in {} are in big-endian, and are in parallel with each other, while the data in {} is in serial with the data in other {}. Data is received from left to right. Take {B0}{B1}{B2}{B3} as an example. The bits in {B0} are in parallel with each other, but are in serial with the bits in {B1}{B2}{B3}. {B0} is received first.

² Only the configuration listed in the table is valid. Other configurations may cause unexpected data errors.

³ B0 ~ B3 represent the bytes of the data to GDMA, from low address to high address.

⁴ In the data to GDMA, Bn'[7:0] = Bn[0:7] (n = 0,1,2,3).

Note:

If only one byte is received each time, CAM_Data_in[7:0] is valid data. For such case, users must connect CAM_Data_in[7:0] with the master.

38.3.6 YUV-RGB Data Format Conversion

LCD_CAM is capable of converting data formats between YUV and RGB. The LCD module and Camera module each have a data format converter. The converters support format conversion:

- under BT601 and BT709 standards
- from RGB565 to YUV422/420/411 (full/limited range)
- from YUV422/YUV411 (full/limited range) to RGB565
- between YUV422/411 (full/limited range) formats

The converter in the LCD module also supports conversion:

- from YUV422/411 (full/limited range) to RGB888 (full/limited range)
- from RGB565 to YUV444 (full/limited range)
- between RGB888 (full/limited range) and RGB565
- from YUV422/411 (full/limited range) to YUV444 (full/limited range)

38.3.6.1 YUV Formats

In LCD_CAM module, assume that there are 8 pixels to be transmitted, corresponding to YUV data $[Y_i, U_i, V_i]$ ($i = 1 \sim 8$). Then:

- In YUV422 mode, the LCD sends (or the camera receives) the data as follows:

Y_1	U_1	Y_2	V_2	Y_3	U_3	Y_4	V_4	Y_5	U_5	Y_6	V_6	Y_7	U_7	Y_8	V_8
-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

- In YUV420 mode, the LCD and the camera do not participate in format conversion.
- In YUV411 mode, the LCD sends (or the camera receives) the data as follows:

Y_1	U_1	Y_2	Y_3	V_3	Y_4	Y_5	U_5	Y_6	Y_7	V_7	Y_8
-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

38.3.6.2 Format Conversion Configuration

The configuration process for the format conversion in the Camera module is almost identical to that in the LCD module. Therefore, we will illustrate the process using the example of format conversion in the LCD module.

1. Enable YUV-RGB format converter by setting [LCD_CAM_LCD_CONV_ENABLE](#).
2. Configure the valid bit width of the input data by configuring [LCD_CAM_LCD_CONV_MODE_8BITS_ON](#):
 - 0: The valid bit width is 16 bits.
 - 1: The valid bit width is 8 bits.
3. Configure the valid bit width of the output data:
 - For the format conversion in the Camera module, the valid bit width of the output data is equal to that of the input data.
 - For the format conversion in the LCD module, when [LCD_CAM_LCD_WIRE_MODE](#) is:
 - 2: The valid bit width of the format converter's output data is 24 bits.
 - Other value: The valid bit width of the format converter's output data is equal to that of the input data.
4. Select the standard by configuring [LCD_CAM_LCD_CONV_PROTOCOL_MODE](#):
 - 0: BT601 standard
 - 1: BT709 standard
5. Configure the conversion mode:
 - In 24-bit output mode:

Table 38.3-5. Conversion Mode Control in 24-bit Mode

Conversion Mode	TRANS_MODE ¹	YUV_MODE ²	YUV2YUV_MODE ³	RGB2RGB_MODE ⁴
RGB565 → YUV444	1	-	3	-
RGB565 → RGB888	0	-	-	1
YUV422 → RGB888	0	0	3	-
YUV411 → RGB888	0	2	3	-
YUV422 → YUV444	1	0	0/1/2	-
YUV411 → YUV444	1	2	0/1/2	-

¹ The value of [LCD_CAM_LCD_CONV_TRANS_MODE](#)² The value of [LCD_CAM_LCD_CONV_YUV_MODE](#)³ The value of [LCD_CAM_LCD_CONV_YUV2YUV_MODE](#)⁴ The value of [LCD_CAM_LCD_CONV_RGB2RGB_MODE](#)

- In other output modes:

Table 38.3-6. Conversion Mode Control in Other Modes

Conversion Mode	TRANS_MODE ¹	YUV_MODE ²	YUV2YUV_MODE ³	RGB2RGB_MODE ⁴
RGB565 → YUV422	1	0	3	-
RGB565 → YUV420	1	1	3	-
RGB565 → YUV411	1	2	3	-
RGB888 → RGB565	0	-	-	1
YUV422 → RGB565	0	0	3	-
YUV411 → RGB565	0	2	3	-
YUV422 → YUV420	1	0	1	-
YUV422 → YUV411	1	0	2	-
YUV411 → YUV422	1	2	0	-
YUV411 → YUV420	1	2	1	-

¹ The value of [LCD_CAM_LCD_CONV_TRANS_MODE](#)² The value of [LCD_CAM_LCD_CONV_YUV_MODE](#)³ The value of [LCD_CAM_LCD_CONV_YUV2YUV_MODE](#)⁴ The value of [LCD_CAM_LCD_CONV_RGB2RGB_MODE](#)

6. Configure the color range for the input data by configuring [LCD_CAM_LCD_CONV_DATA_IN_MODE](#):

- 0: limited color range¹
- 1: full color range²

7. Configure the color range for the output data by configuring [LCD_CAM_LCD_CONV_DATA_OUT_MODE](#):

- 0: limited color range¹
- 1: full color range²

Note:

1. If the limited color range is selected,

- the color range of RGB is: 16 ~ 240.
- the color range of YUV is:
 - Y: 16 ~ 240.
 - U-V: 16 ~ 235.

2. If the full color range is selected, the color range of RGB or YUV is 0 ~ 255.

38.3.7 LCD_CAM Timing

38.3.7.1 LCD Timing (RGB Format)

Figure 38.3-4 shows the LCD frame structure.

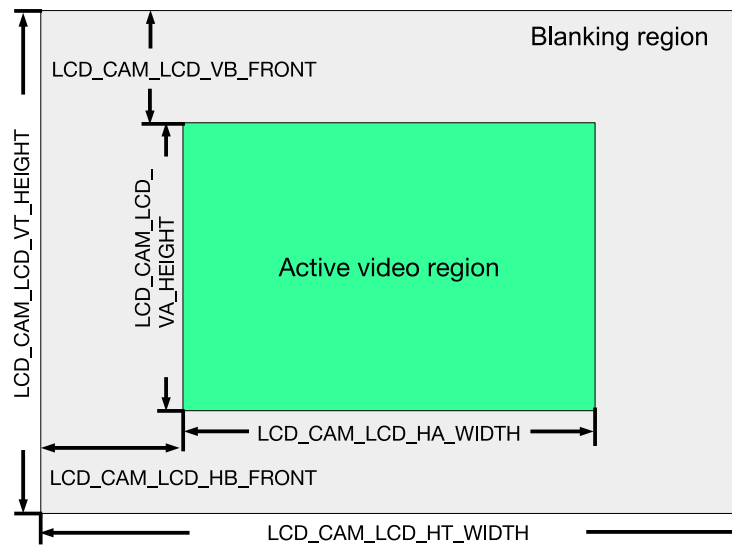


Figure 38.3-4. LCD Frame Structure

As depicted in the figure, the frame structure can be set up with the following registers:

- [LCD_CAM_LCD_VT_HEIGHT](#)
- [LCD_CAM_LCD_VA_HEIGHT](#)
- [LCD_CAM_LCD_HB_FRONT](#)
- [LCD_CAM_LCD_HT_WIDTH](#)
- [LCD_CAM_LCD_HA_WIDTH](#)
- [LCD_CAM_VB_FRONT](#)

Figure 38.3-5 shows the timing of an LCD full frame.

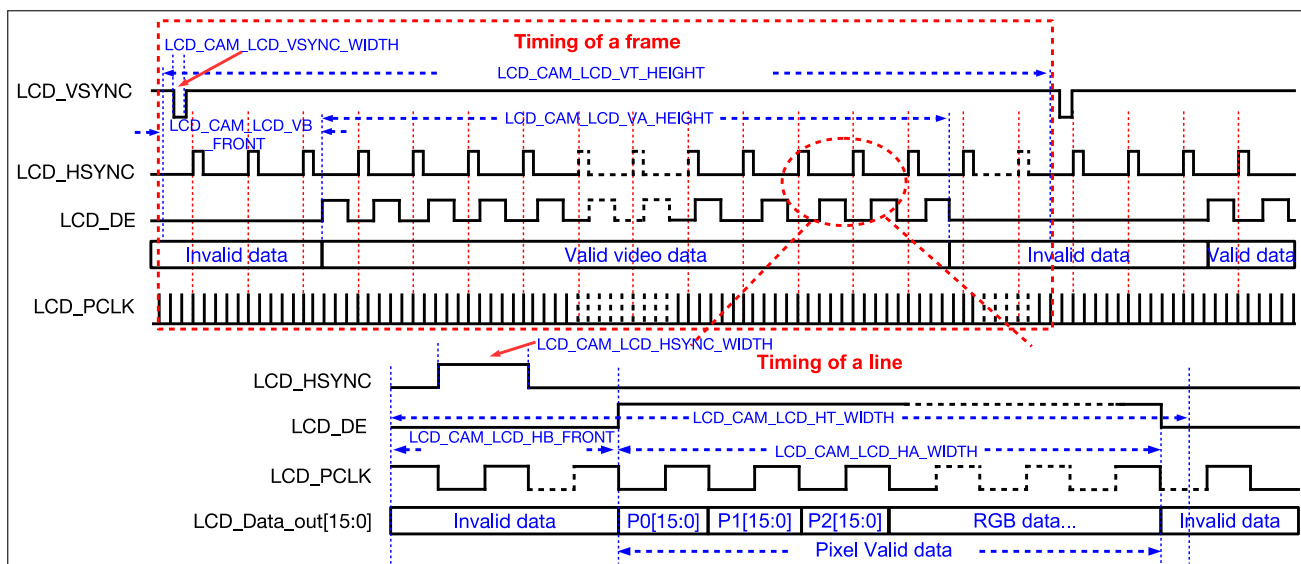


Figure 38.3-5. LCD Timing (RGB Format)

Note:

When configuring the parameters shown in the figures above, please note that the parameter is equal to the register value + 1. For example, if the expected VSYNC width is 1, then please configure `LCD_CAM_LCD_VSYNC_WIDTH` to 0. For more information, see the register description.

38.3.7.2 LCD Timing (I8080/MOTO6800 Format)

Figure 38.3-6 shows the LCD timing sequence in I8080/MOTO6800 format.

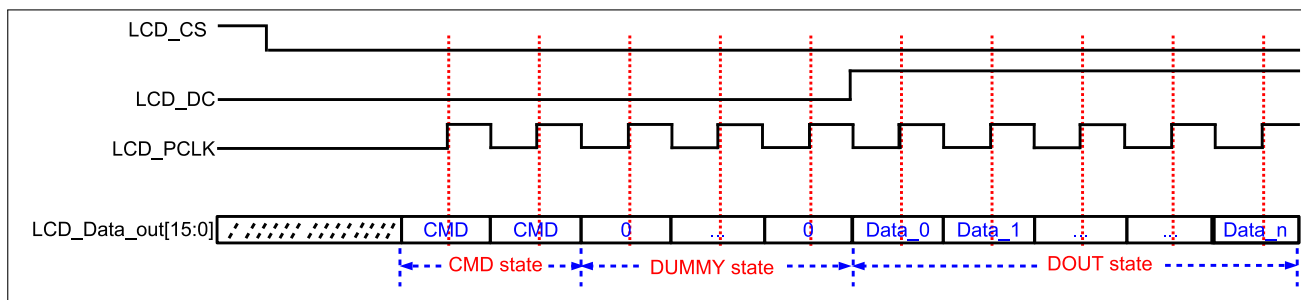


Figure 38.3-6. LCD Timing (I8080/MOTO6800 Format)

38.4 Interrupts

ESP32-P4's LCD_CAM can generate the following interrupt signal that will be sent to the [Interrupt Matrix](#).

- `LCD_CAM_INTR`

The following internal interrupt sources from LCD_CAM can generate `LCD_CAM_INTR`:

- `LCD_CAM_CAM_HS_INT`: Triggered when the Camera module receives lines more than or equal to the value of `LCD_CAM_CAM_LINE_INT_NUM + 1`.
- `LCD_CAM_CAM_VSYNC_INT`: Triggered when the Camera module receives a whole frame.

- LCD_CAM_LCD_TRANS_DONE_INT: Triggered when the LCD module finishes the transmission.
- LCD_CAM_LCD_VSYNC_INT: Triggered when the LCD module transmits a whole frame.

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

38.5 Software Configuration Process

Note:

Updating the configurations of the LCD module or the Camera module requires setting the LCD_CAM_LCD_UPDATE and LCD_CAM_CAM_UPDATE respectively to synchronize registers from the APB clock domain to the LCD/camera clock domain. Check the configuration examples below for more details.

38.5.1 Configure LCD (RGB Format) as TX Mode

Follow the steps below to configure LCD (RGB format) as TX mode via software:

1. Configure clock according to Section [38.3.3](#).
2. Configure signal pins according to Table [38.3-1](#) and Chapter [9 GPIO Matrix and IO MUX](#).
3. Enable corresponding interrupts, see Section [38.4](#).
4. Enable RGB format by setting [LCD_CAM_LCD_RGB_MODE_EN](#).
5. Configure frame structure according to Section [38.3.71](#).
6. Set [LCD_CAM_LCD_UPDATE](#) so that the configurations take effect.
7. Reset TX control unit (LCD_Ctrl) and Async TX FIFO as described in Section [38.3.4](#).
8. Configure GDMA outlink, see Chapter [5 VDMA Controller \(VDMA\)](#).
9. Start transmitting data:
 - Wait till LCD slave gets ready.
 - Set [LCD_CAM_LCD_START](#) to start transmitting data.
10. Wait for the interrupt signals set in Step [3](#).
11. In frame intervals during data transmission, check if [LCD_CAM_LCD_NEXT_FRAME_EN](#) is set:
 - If yes, set [LCD_CAM_LCD_UPDATE](#) (repeating the steps above) to continue transmitting next frame.
 - If not, LCD stops transmitting data.
12. Clear [LCD_CAM_LCD_START](#) if data transmission is done.

38.5.2 Configure LCD (I8080/MOTO6800 Format) as TX Mode

Follow the steps below to configure LCD (I8080/MOTO6800 format) as TX mode via software:

1. Configure clock according to Section [38.3.3](#).

2. Configure signal pins according to Table [38.3-1](#) and Chapter [9 GPIO Matrix and IO MUX](#).
3. Enable corresponding interrupts, see Section [38.4](#).
4. Clear [LCD_CAM_LCD_RGB_MODE_EN](#) to disable RGB format.
5. Configure CMD phase with [LCD_CAM_LCD_CMD](#), [LCD_CAM_LCD_CMD_2_CYCLE_EN](#), and [LCD_CAM_LCD_LATTER_CMD_VALUE](#).
6. Configure DUMMY phase with [LCD_CAM_LCD_DUMMY](#) and [LCD_CAM_LCD_DUMMY_CYCLELEN](#).
7. Configure DOUT phase with [LCD_CAM_LCD_DOUT](#), depending on the output mode:
 - In a fixed-length output¹, configure data length in [LCD_CAM_LCD_DOUT_CYCLELEN](#).
 - In a continuous output², set [LCD_CAM_LCD_ALWAYS_OUT_EN](#). Users do not need to configure [LCD_CAM_LCD_DOUT_CYCLELEN](#).

Note:

- (a) In a fixed-length output, the LCD module stops sending data once the data length reaches the value set in [LCD_CAM_LCD_DOUT_CYCLELEN](#).
- (b) In a continuous output, LCD module keeps sending data till:
 - [LCD_CAM_LCD_START](#) is cleared;
 - or [LCD_CAM_LCD_RESET](#) is set;
 - or all the data in GDMA is sent out.

8. Configure the CD signal mode, including the default value of the CD signal and the values at each phase, see the description of [LCD_CAM_LCD_MISC_REG](#).
9. Set [LCD_CAM_LCD_UPDATE](#) so that the configurations take effect.
10. Reset TX control unit (LCD_Ctrl) and Async TX FIFO as described in Section [38.3.4](#).
11. Configure GDMA outlink, see Chapter [5 VDMA Controller \(VDMA\)](#).
12. Start transmitting data:
 - Wait till LCD slave gets ready.
 - Set [LCD_CAM_LCD_START](#) to start transmitting data.
13. Wait for the interrupt signals set in Step 2.
14. Clear [LCD_CAM_LCD_START](#) if data transmission is done.

Notes:

No matter in which format, RGB or I8080/MOTO6800, the rules below must be followed when accessing internal and external memory via GDMA:

- If LCD data bus is configured to 8-bit parallel output mode, then
 - The pixel clock frequency must be less than 80 MHz.
 - If YUV-RGB format conversion is being used at the same time, the pixel clock frequency must be less than 60 MHz.
- If LCD data bus is configured to 16-bit parallel output mode, then

- The pixel clock frequency must be less than 40 MHz.
- If YUV-RGB format conversion is being used at the same time, the pixel clock frequency must be less than 30 MHz.

38.5.3 Configure Camera as RX Mode

Follow the steps below to configure the Camera as RX mode via software:

1. Configure clock according to Section 38.3.3. Note that in slave mode, the module clock frequency should be twice as fast as the PCLK frequency of the image sensor.
2. Configure signal pins according to Table 38.3-1 and Chapter 9 *GPIO Matrix and IO MUX*.
3. Set or clear `LCD_CAM_CAM_VH_DE_MODE_EN` according to the control signal HSYNC.
4. Set needed RX data format according to Section 38.3.5.2.
5. Set `LCD_CAM_CAM_UPDATE` so that the configurations take effect.
6. Reset RX control unit (Camera_Ctrl) and Async RX FIFO as described in Section 38.3.4.
7. Enable corresponding interrupts, see Section 38.4.
8. Configure GDMA inlink, and set the length of RX data in `LCD_CAM_CAM_REC_DATA_BYTELEN`.
9. Start receiving data:
 - In master mode, when the slave is ready, set `LCD_CAM_CAM_START` to start receiving data.
 - In slave mode, set `LCD_CAM_CAM_START`. Receiving data starts after the master provides clock signal and control signal.
10. Receive data and store the data to GDMA. Then corresponding interrupts set in Step 7 will be generated.

Notes:

- No matter in which operation mode, camera master RX mode or camera slave RX mode, the rules below must be followed when accessing internal memory via GDMA:
 - If 8-bit parallel data input mode is selected, then
 - * The pixel clock frequency must be less than 80 MHz.
 - * If YUV-RGB format is being used at the same time, the pixel clock frequency must be less than 60 MHz.
 - If 16-bit parallel data input mode is selected, then
 - * The pixel clock frequency must be less than 40 MHz.
 - * If YUV-RGB format conversion is being used at the same time, the pixel clock frequency must be less than 30 MHz.

Note:

If both LCD and camera are connected externally at the same time, it is necessary to ensure that the maximum data throughput on the interface is less than the total data bandwidth of GDMA when accessing internal/external storage. The default frequency of APB_CLK is 80 MHz in this scenario. For more information about APB_CLK, see Chapter 10 [Reset and Clock](#).

38.6 Register Summary

The addresses in this section are relative to LCD_CAM base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
LCD Module Configuration Registers			
LCD_CAM_LCD_CLOCK_REG	LCD clock configuration register	0x0000	R/W
LCD_CAM_LCD_RGB_YUV_REG	LCD data format conversion register	0x0010	R/W
LCD_CAM_LCD_USER_REG	LCD user configuration register	0x0014	varies
LCD_CAM_LCD_MISC_REG	LCD MISC configuration register	0x0018	varies
LCD_CAM_LCD_CTRL_REG	LCD signal configuration register	0x001C	R/W
LCD_CAM_LCD_CTRL1_REG	LCD signal configuration register 1	0x0020	R/W
LCD_CAM_LCD_CTRL2_REG	LCD signal configuration register 2	0x0024	R/W
LCD_CAM_LCD_FIRST_CMD_VAL_REG	LCD command value configuration register	0x0028	R/W
LCD_CAM_LCD_LATTER_CMD_VAL_REG	LCD command value configuration register	0x002C	R/W
LCD_CAM_LCD_DLY_MODE_CFG1_REG	LCD data/signal delay configuration register	0x0030	R/W
LCD_CAM_LCD_DLY_MODE_CFG2_REG	LCD signal delay mode configuration register	0x0038	R/W
Camera Module Configuration Registers			
LCD_CAM_CAM_CTRL_REG	Camera clock configuration	0x0004	R/W
LCD_CAM_CAM_CTRL1_REG	Camera control register	0x0008	varies
LCD_CAM_CAM_RGB_YUV_REG	Camera data format conversion register	0x000C	R/W
Interrupt Registers			
LCD_CAM_LC_DMA_INT_ENA_REG	LCD_CAM GDMA interrupt enable register	0x0064	R/W
LCD_CAM_LC_DMA_INT_RAW_REG	LCD_CAM GDMA raw interrupt status register	0x0068	RO
LCD_CAM_LC_DMA_INT_ST_REG	LCD_CAM GDMA masked interrupt status register	0x006C	RO
LCD_CAM_LC_DMA_INT_CLR_REG	LCD_CAM GDMA interrupt clear register	0x0070	WO
Version Control Register			
LCD_CAM_LC_DATE_REG	Version control register	0x00FC	R/W

38.7 Registers

The addresses in this section are relative to LCD_CAM base address provided in Table 7.3-2 in Chapter 7 [System and Memory](#).

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 38.1. LCD_CAM_LCD_CLOCK_REG (0x0000)

LCD_CAM_CLK_EN		(reserved)							LCD_CAM_LCD_CLK_CNT_N		LCD_CAM_LCD_CLK_EQU_SYSCLK		LCD_CAM_LCD_CLK_IDLE_EDGE		LCD_CAM_LCD_CLK_OUT_EDGE	
31	30								9	8	7	6	5			0
0	0x0								0	0	1	0x3				Reset

LCD_CAM_LCD_CLKCNT_N Determines the frequency of the LCD pixel clock LCD_PCLK together with [LCD_CAM_LCD_CLK_EQU_SYSCLK](#).

When [LCD_CAM_LCD_CLK_EQU_SYSCLK](#) = 0,

$$f_{LCD_PCLK} = f_{LCD_CLK} / (LCD_CAM_LCD_CLKCNT_N + 1)$$

Note: this field must not be configured to 0.

(R/W)

LCD_CAM_LCD_CLK_EQU_SYSCLK Determines the frequency of the LCD pixel clock LCD_PCLK together with [LCD_CAM_LCD_CLK_EQU_SYSCLK](#).

$$0: f_{LCD_PCLK} = f_{LCD_CLK} / (LCD_CAM_LCD_CLKCNT_N + 1)$$

$$1: f_{LCD_PCLK} = f_{LCD_CLK}$$

(R/W)

LCD_CAM_LCD_CLK_IDLE_EDGE Indicates the level of LCD_PCLK in idle state.

0: LCD_PCLK is low.

1: LCD_PCLK is high.

(R/W)

LCD_CAM_LCD_CLK_OUT_EDGE Indicates the level of LCD_PCLK in the first half clock cycle.

0: LCD_PCLK is low.

1: LCD_PCLK is high.

(R/W)

LCD_CAM_CLK_EN Set this bit to force enable the clock for all configuration registers. Clock gate is not used. (R/W)

LCD_CAM_LCD_CONV_ENABLE
 LCD_CAM_LCD_CONV_TRANS_MODE
 LCD_CAM_LCD_CONV_MODE_8BITS_ON
 LCD_CAM_LCD_CONV_DATA_IN_MODE
 LCD_CAM_LCD_CONV_DATA_OUT_MODE
 LCD_CAM_LCD_CONV_PROTOCOL_MODE
 LCD_CAM_LCD_CONV_YUV_MODE
 LCD_CAM_LCD_CONV_YUV2YUV_MODE
 LCD_CAM_LCD_CONV_8BITS_DATA_INV
 LCD_CAM_LCD_CONV_RGB2RGB_MODE
 (reserved)

Reset

(R/W)

(R/W)

(R/W)

(R/W)

(R/W)

(R/W)

ESP32-P4 TRM
PRELIMINARY

Register 38.2. LCD_CAM_LCD_RGB_YUV_REG (0x0010)

Continued from the previous page...

LCD_CAM_LCD_CONV_DATA_IN_MODE Configures the color range for input data.

0: Limited color range.

1: Full color range.

(R/W)

LCD_CAM_LCD_CONV_MODE_8BITS_ON Configures the valid bit width of the YUV-RGB converter's input data.

0: 16 bits.

1: 8 bits.

(R/W)

LCD_CAM_LCD_CONV_TRANS_MODE Configures the RGB or YUV format for data conversion.

0: Converted to RGB format.

1: Converted to YUV format.

(R/W)

LCD_CAM_LCD_CONV_ENABLE Configures whether to enable YUV-RGB converter.

0: Bypass the converter.

1: Enable the converter.

(R/W)

Register 38.3. LCD_CAM_LCD_USER_REG (0x0014)

<div style="display: flex; justify-content: space-between;"> <div> <div>LCD_CAM_LCD_CMD_2_CYCLE_EN</div> <div>LCD_CAM_LCD_DUMMY_CYCLELEN</div> <div>LCD_CAM_LCD_RESET</div> <div>LCD_CAM_LCD_START</div> <div>LCD_CAM_LCD_CMD</div> <div>LCD_CAM_LCD_DUMMY</div> <div>LCD_CAM_LCD_DOUT</div> <div>LCD_CAM_LCD_BYTE_ORDER</div> <div>LCD_CAM_LCD_BIT_ORDER</div> <div>LCD_CAM_LCD_UPDATE</div> <div>LCD_CAM_LCD_BYTE_MODE</div> <div>LCD_CAM_LCD_DOUT_BIT_ORDER</div> <div>LCD_CAM_LCD_DOUT_BYTE_SWIZZLE_ENABLE</div> <div>LCD_CAM_LCD_DOUT_BYTE_SWIZZLE_MODE</div> <div>LCD_CAM_LCD_ALWAYS_OUT_EN</div> <div>LCD_CAM_LCD_DOUT_CYCLELEN</div> </div> <div> <div>31</div><div>30</div><div>29</div><div>28</div><div>27</div><div>26</div><div>25</div><div>24</div><div>23</div><div>22</div><div>21</div><div>20</div><div>19</div><div>18</div><div>17</div><div>16</div><div>14</div><div>13</div><div>12</div><div data-cs="12" data-kind="parent"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div data-kind="ghost"></div><div>0</div> </div> </div>																																					
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x01																		Reset

LCD_CAM_LCD_DOUT_CYCLELEN Configures the period of the LCD module's data output phase (DOUT). The actual period duration = the value of this field + 1. (R/W)

LCD_CAM_LCD_ALWAYS_OUT_EN Enables continuous output of the LCD module. In this mode, the LCD module continuously outputs data during the DOUT phase, until [LCD_CAM_LCD_START](#) is cleared or [LCD_CAM_LCD_RESET](#) is set. (R/W)

LCD_CAM_LCD_DOUT_BYTE_SWIZZLE_MODE Configures LCD output data byte reordering. For details please refer to Table [38.3-3](#). (R/W)

LCD_CAM_LCD_DOUT_BYTE_SWIZZLE_ENABLE Configures whether to enable LCD output data byte reordering.
 0: Disable
 1: Enable
 (R/W)

LCD_CAM_LCD_DOUT_BIT_ORDER Configures whether to invert the LCD output data bit order.
 0: Do not invert.
 1: Invert the bit order. In n-bit mode, LCD_DATA_in[n-1:0] is inverted to LCD_DATA_in[0:n-1].
 (R/W)

LCD_CAM_LCD_BYTE_MODE Configures the data bit width from GDMA.
 0: 8-bit
 1: 16-bit
 2: 24-bit
 (R/W)

LCD_CAM_LCD_UPDATE Configures whether to update the LCD register configurations.
 0: Do not update.
 1: Update configurations. This bit is cleared by hardware.
 (R/W)

LCD_CAM_LCD_BIT_ORDER Configures whether to invert the bit order of the LCD input data.
 0: Do not invert.
 1: Invert the bit order. In n-bit mode, LCD_DATA_in[n-1:0] is inverted to LCD_DATA_in[0:n-1].
 (R/W)

Continued on the next page...

Register 38.3. LCD_CAM_LCD_USER_REG (0x0014)

Continued from the previous page...

LCD_CAM_LCD_BYTE_ORDER Configures whether to invert the byte order of the LCD input data.

0: Do not invert.

1: Invert data byte order, only valid in 16/24-bit mode.

(R/W)

LCD_CAM_LCD_DOUT Configures the LCD DOUT phase.

0: Disable.

1: The LCD module sends data in RGB/I8080 format.

(R/W)

LCD_CAM_LCD_DUMMY Configures the LCD DUMMY phase.

0: Disable.

1: Enable the DUMMY phase when LCD initiates.

(R/W)

LCD_CAM_LCD_CMD Configures the LCD CMD phase.

0: Disable.

1: The LCD module sends commands.

(R/W)

LCD_CAM_LCD_START When set to 1, the LCD module starts transmitting data. (R/W)

LCD_CAM_LCD_RESET When set to 1, the LCD module is reset. (WO)

LCD_CAM_LCD_DUMMY_CYCLELEN Configures DUMMY cycles. DUMMY cycles = this value + 1.

(R/W)

LCD_CAM_LCD_CMD_2_CYCLE_EN Configures the number of cycles of the Command phase.

0: One cycle.

1: Two cycles.

(R/W)

Register 38.4. LCD_CAM_LCD_MISC_REG (0x0018)

LCD_CAM_LCD_CD_IDLE_EDGE LCD_CAM_LCD_CD_CMD_SET LCD_CAM_LCD_CD_DUMMY_SET LCD_CAM_LCD_CD_DATA_SET LCD_CAM_LCD_AFIFO_RESET LCD_CAM_LCD_BK_EN LCD_CAM_LCD_NEXT_FRAME_EN								LCD_CAM_LCD_VBK_CYCLELEN				LCD_CAM_LCD_VFK_CYCLELEN				LCD_CAM_LCD_WIRE_MODE (reserved)			
31	30	29	28	27	26	25	24	12				11	6		5	4	3	0	
0	0	0	0	0	0	0	0x00				0x3		0		0		Reset		

Reset

LCD_CAM_LCD_WIRE_MODE Configures the bit width of LCD output data to GPIO.

0: 8-bit

1: 16-bit

2: 24-bit

(R/W)

LCD_CAM_LCD_VFK_CYCLELEN Configures the clock cycles of the setup time in LCD non-RGB mode. Setup clock cycles = this value + 1. (R/W)**LCD_CAM_LCD_VBK_CYCLELEN** Configures the clock cycles of the hold time in LCD non-RGB mode. Hold clock cycles = this value + 1. (R/W)**LCD_CAM_LCD_NEXT_FRAME_EN** Configures whether to send the next frame.

0: The LCD module stops when the current frame is sent out.

1: The LCD module continues sending the next frame when the current frame is sent out.

(R/W)

LCD_CAM_LCD_BK_EN Configures whether to enable blanking region when LCD sends data.

0: No blanking region.

1: Enable blanking region.

(R/W)

LCD_CAM_LCD_AFIFO_RESET When set to 1, Async TX FIFO is reset. (WO)**LCD_CAM_LCD_CD_DATA_SET** Configures LCD_CD value in DOUT phase for LCD I8080 mode.0: LCD_CD = [LCD_CAM_LCD_CD_IDLE_EDGE](#)1: LCD_CD = [!LCD_CAM_LCD_CD_IDLE_EDGE](#)

(R/W)

LCD_CAM_LCD_CD_DUMMY_SET Configures LCD_CD value in DUMMY phase for LCD I8080 mode.0: LCD_CD = [LCD_CAM_LCD_CD_IDLE_EDGE](#)1: LCD_CD = [!LCD_CAM_LCD_CD_IDLE_EDGE](#)

(R/W)

LCD_CAM_LCD_CD_CMD_SET Configures LCD_CD value in CMD phase for LCD I8080 mode.0: LCD_CD = [LCD_CAM_LCD_CD_IDLE_EDGE](#)1: LCD_CD = [!LCD_CAM_LCD_CD_IDLE_EDGE](#)

(R/W)

LCD_CAM_LCD_CD_IDLE_EDGE The default value of LCD_CD. (R/W)

Register 38.5. LCD_CAM_LCD_CTRL_REG (0x001C)

LCD_CAM_LCD_RGB_MODE_EN																															LCD_CAM_LCD_VT_HEIGHT																															LCD_CAM_LCD_VA_HEIGHT																															LCD_CAM_LCD_HB_FRONT																														
31	30																														21										20	11										10	0																																																																						
0	0										0										0										0										Reset																																																																																		

LCD_CAM_LCD_HB_FRONT Configures the value of ([HSYNC_POSITION](#) + [HSYNC_WIDTH](#) + horizontal back porch). (R/W)

LCD_CAM_LCD_VA_HEIGHT Configures the vertical active height of a frame. (R/W)

LCD_CAM_LCD_VT_HEIGHT Configures the vertical total height of a frame. (R/W)

LCD_CAM_LCD_RGB_MODE_EN Configures whether to enable RGB mode.

0: Disable RGB mode.

1: Enable RGB mode and input VSYNC, HSYNC, and DE signals.

(R/W)

Register 38.6. LCD_CAM_LCD_CTRL1_REG (0x0020)

LCD_CAM_LCD_HT_WIDTH										LCD_CAM_LCD_HA_WIDTH										LCD_CAM_LCD_VB_FRONT														
31										20	19									8	7													0
0										0										0												Reset		

LCD_CAM_LCD_VB_FRONT Configures the value of ([VSYNC_WIDTH](#) + vertical back porch). (R/W)

LCD_CAM_LCD_HA_WIDTH Configures the horizontal active width of a frame. (R/W)

LCD_CAM_LCD_HT_WIDTH Configures the horizontal total width of a frame. (R/W)

Register 38.7. LCD_CAM_LCD_CTRL2_REG (0x0024)

LCD_CAM_LCD_HSYNC_POSITION																
LCD_CAM_LCD_HSYNC_IDLE_POL																
LCD_CAM_LCD_HSYNC_WIDTH																
(reserved)																
LCD_CAM_LCD_HS_BLANK_EN																
LCD_CAM_LCD_DE_IDLE_POL																
LCD_CAM_LCD_VSYNC_IDLE_POL																
LCD_CAM_LCD_VSYNC_WIDTH																
31	24		23	22		16		15	10		9	8	7	6	0	
0			0	1		0		0	0	0	0	0	0	0	1	
Reset																

Reset

LCD_CAM_LCD_VSYNC_WIDTH Configures the width of LCD_VSYNC active pulse in a line. Expected width = this value + 1. (R/W)

LCD_CAM_LCD_VSYNC_IDLE_POL Configures the idle value of LCD_VSYNC.

0: The idle value is low.

1: The idle value is high.

(R/W)

LCD_CAM_LCD_DE_IDLE_POL Configures the idle value of LCD_DE.

0: The idle value is low.

1: The idle value is high.

(R/W)

LCD_CAM_LCD_HS_BLANK_EN Configures the LCD_HSYNC output in RGB mode.

0: LCD_HSYNC is output only in active video lines.

1: LCD_HSYNC can be output in vertical blanking intervals.

(R/W)

LCD_CAM_LCD_HSYNC_WIDTH Configures the width of LCD_HSYNC active pulse. Expected width = this value +1. (R/W)

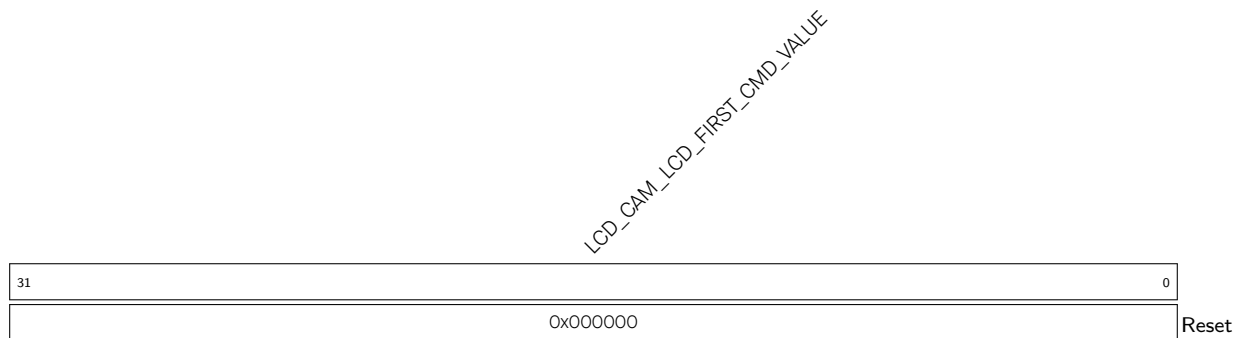
LCD_CAM_LCD_HSYNC_IDLE_POL Configures the idle value of LCD_HSYNC.

0: The idle value is low.

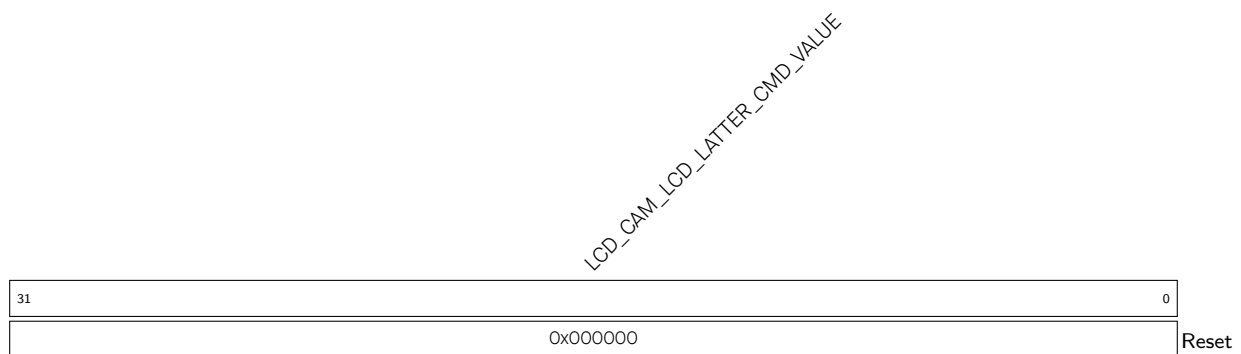
1: The idle value is high.

(R/W)

LCD_CAM_LCD_HSYNC_POSITION Configures the position of LCD_HSYNC active pulse. Expected position = this value + 1. Unit is a pixel. (R/W)

Register 38.8. LCD_CAM_LCD_FIRST_CMD_VAL_REG (0x0028)

LCD_CAM_LCD_FIRST_CMD_VALUE Configures the value of the transmitted command when the LCD module is in the first cycle of the CMD phase. (R/W)

Register 38.9. LCD_CAM_LCD_LATTER_CMD_VAL_REG (0x002C)

LCD_CAM_LCD_LATTER_CMD_VALUE Configures the value of the transmitted command when the LCD module is in the second cycle of the CMD phase. (R/W)

Register 38.10. LCD_CAM_LCD_DLY_MODE_CFG1_REG (0x0030)

(reserved)								LCD_CAM_LCD_VSYNC_MODE		LCD_CAM_LCD_HSYNC_MODE		LCD_CAM_LCD_DE_MODE		LCD_CAM_LCD_CD_MODE		LCD_CAM_DOUT23_MODE		LCD_CAM_DOUT22_MODE		LCD_CAM_DOUT21_MODE		LCD_CAM_DOUT20_MODE		LCD_CAM_DOUT19_MODE		LCD_CAM_DOUT18_MODE		LCD_CAM_DOUT17_MODE		LCD_CAM_DOUT16_MODE	
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	0x0	Reset

LCD_CAM_DOUT n _MODE (n : 16 - 23) Configures the delay of the output data bit (n).

- 0: No delay.
 - 1: Delayed at the rising edge of LCD_CLK.
 - 2: Delayed at the falling edge of LCD_CLK.
- (R/W)

LCD_CAM_LCD_CD_MODE Configures the delay of LCD_CD output signal.

- 0: No delay.
 - 1: Delayed at the rising edge of LCD_CLK.
 - 2: Delayed at the falling edge of LCD_CLK.
- (R/W)

LCD_CAM_LCD_DE_MODE Configures the delay of LCD_DE output signal.

- 0: No delay.
 - 1: Delayed at the rising edge of LCD_CLK.
 - 2: Delayed at the falling edge of LCD_CLK.
- (R/W)

LCD_CAM_LCD_HSYNC_MODE Configures the delay of LCD_HSYNC output signal.

- 0: No delay.
 - 1: Delayed at the rising edge of LCD_CLK.
 - 2: Delayed at the falling edge of LCD_CLK.
- (R/W)

LCD_CAM_LCD_VSYNC_MODE Configures the delay of LCD_VSYNC output signal.

- 0: No delay.
 - 1: Delayed at the rising edge of LCD_CLK.
 - 2: Delayed at the falling edge of LCD_CLK.
- (R/W)

Register 38.11. LCD_CAM_LCD_DLY_MODE_CFG2_REG (0x0038)

LCD_CAM_DOUT15_MODE		LCD_CAM_DOUT14_MODE		LCD_CAM_DOUT13_MODE		LCD_CAM_DOUT12_MODE		LCD_CAM_DOUT11_MODE		LCD_CAM_DOUT10_MODE		LCD_CAM_DOUT9_MODE		LCD_CAM_DOUT8_MODE		LCD_CAM_DOUT7_MODE		LCD_CAM_DOUT6_MODE		LCD_CAM_DOUT5_MODE		LCD_CAM_DOUT4_MODE		LCD_CAM_DOUT3_MODE		LCD_CAM_DOUT2_MODE		LCD_CAM_DOUT1_MODE		LCD_CAM_DOUT0_MODE	
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		0x0		Reset	

LCD_CAM_DOUT_{*n*}_MODE (*n*: 0 - 15) Configures the delay of the output data bit *n*.

- 0: No delay.
- 1: Delayed at the rising edge of LCD_CLK.
- 2: Delayed at the falling edge of LCD_CLK.

(R/W)

Register 38.12. LCD_CAM_CAM_CTRL_REG (0x0004)

(reserved)										LCD_CAM_CAM_VS_EOF_EN								
										LCD_CAM_CAM_LINE_INT_EN								
										LCD_CAM_CAM_BIT_ORDER								
										LCD_CAM_CAM_BYTE_ORDER								
										LCD_CAM_CAM_UPDATE								
										LCD_CAM_CAM_VSYNC_FILTER_THRES								
										LCD_CAM_CAM_STOP_EN								
31									9	8	7	6	5	4	3	1		0
0x0										0	0	0	0	0	0x0		0	Reset

LCD_CAM_CAM_STOP_EN Configures whether to stop the Camera module.

0: Do not stop.

1: The Camera module stops when GDMA RX FIFO is full.

(R/W)

LCD_CAM_CAM_VSYNC_FILTER_THRES Configures the filter threshold value for CAM_VSYNC signal. (R/W)

LCD_CAM_CAM_UPDATE Configures whether to update the Camera register configurations.

0: No effect.

1: Update configurations. This bit is cleared by hardware.

(R/W)

LCD_CAM_CAM_BYTE_ORDER Configures whether to invert the byte order of the Camera's input data.

0: Do not invert.

1: Invert data byte order, only valid in 16-bit mode.

(R/W)

LCD_CAM_CAM_BIT_ORDER Configures whether to invert the bit order of the Camera's input data.

0: Do not invert.

1: Invert the bit order. In n-bit mode, CAM_DATA_in[n-1:0] is inverted to CAM_DATA_in[0:n-1]. n = 8 or 15.

(R/W)

LCD_CAM_CAM_LINE_INT_EN Configures whether the RAW bit of [LCD_CAM_CAM_HS_INT](#) can generate.

0: Disable the interrupt, i.e., [LCD_CAM_CAM_HS_INT_RAW](#) cannot generate.

1: Enable the interrupt, i.e., [LCD_CAM_CAM_HS_INT_RAW](#) can generate.

(R/W)

LCD_CAM_CAM_VS_EOF_EN Configures whether to enable CAM_VSYNC to generate in_suc_eof.

0: in_suc_eof is controlled by [LCD_CAM_CAM_REC_DATA_BYTELEN](#).

1: in_suc_eof is controlled by CAM_VSYNC.

(R/W)

LCD_CAM_CAM_AFIFO_RESET
LCD_CAM_CAM_RESET
LCD_CAM_CAM_START
LCD_CAM_CAM_VH_DE_MODE_EN
LCD_CAM_CAM_VSYNC_INV
LCD_CAM_CAM_DE_INV
LCD_CAM_CAM_2BYTE_EN
LCD_CAM_CAM_CLK_INV
LCD_CAM_CAM_LINE_INT_NUM
LCD_CAM_CAM_REC_DATA_BYTELEN

Reset

LCD_CAM_CAM_VSYNC_INV Configures whether to invert the input signal CAM_VSYNC.
0: Do not invert.
1: Invert.
(R/W)

Continued on the next page...

Register 38.13. LCD_CAM_CAM_CTRL1_REG (0x0008)

Continued from the previous page...

LCD_CAM_CAM_VH_DE_MODE_EN Configures the input control signals.

0: VSYNC and DE signals control the data. In this case, wiring HSYNC signal line is not a must. But in this case, the YUV-RGB conversion function of the camera module is not available.

1: VSYNC, HSYNC, and DE signals control the data. In this case, users need to wire the three signal lines.

(R/W)

LCD_CAM_CAM_START Camera module start signal. (R/W)

LCD_CAM_CAM_RESET When set to 1, the Camera module is reset. (WO)

LCD_CAM_CAM_AFIFO_RESET When set to 1, the Camera Async RX FIFO is reset. (WO)

Register 38.14. LCD_CAM_CAM_RGB_YUV_REG (0x000C)

LCD_CAM_CAM_CONV_ENABLE LCD_CAM_CAM_CONV_TRANS_MODE LCD_CAM_CAM_CONV_MODE_8BITS_ON LCD_CAM_CAM_CONV_DATA_IN_MODE LCD_CAM_CAM_CONV_DATA_OUT_MODE LCD_CAM_CAM_CONV_PROTOCOL_MODE LCD_CAM_CAM_CONV_YUV_MODE LCD_CAM_CAM_CONV_YUV2YUV_MODE LCD_CAM_CAM_CONV_8BITS_DATA_INV (reserved)																															
31	30	29	28	27	26	25	24	23	22	21	20																				0
0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

LCD_CAM_CAM_CONV_8BITS_DATA_INV Configures whether to swap every two bytes of input data.

0: Do not invert.

1: Invert.

(R/W)

LCD_CAM_CAM_CONV_YUV2YUV_MODE Configures the data format conversion in YUV-to-YUV mode.

0: Data is converted to YUV422 format.

1: Data is converted to YUV420 format.

2: Data is converted to YUV411 format.

3: Disabled.

Valid only when [LCD_CAM_CAM_CONV_TRANS_MODE](#) = 1.

(R/W)

LCD_CAM_CAM_CONV_YUV_MODE Configures the YUV format of the Camera input data in YUV-to-YUV mode or YUV-to-RGB mode.

0: Data is converted to YUV422 format.

1: Data is converted to YUV420 format.

2: Data is converted to YUV411 format.

(R/W)

LCD_CAM_CAM_CONV_PROTOCOL_MODE Configures the data format conversion standard.

0: BT601.

1: BT709.

(R/W)

LCD_CAM_CAM_CONV_DATA_OUT_MODE Configures the color range for the YUV-RGB converter's output data.

0: Limited color range.

1: Full color range.

(R/W)

Continued on the next page...

Register 38.14. LCD_CAM_CAM_RGB_YUV_REG (0x000C)

Continued from the previous page...

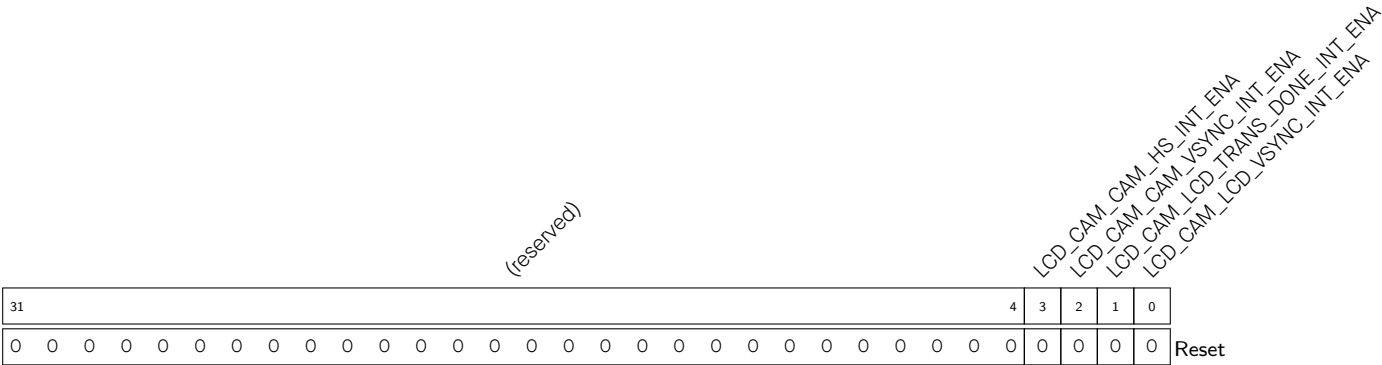
LCD_CAM_CAM_CONV_DATA_IN_MODE Configures the color range for the YUV-RGB converter's input data.
0: Limited color range.
1: Full color range.
(R/W)

LCD_CAM_CAM_CONV_MODE_8BITS_ON Configures the bit width of the YUV-RGB converter's input data.
0: 16-bit mode.
1: 8-bit mode.
(R/W)

LCD_CAM_CAM_CONV_TRANS_MODE Configures the data conversion format.
0: Data is converted to RGB format.
1: Data is converted to YUV format.
(R/W)

LCD_CAM_CAM_CONV_ENABLE Configures whether to enable the YUV-RGB converter.
0: Bypass.
1: Enable.
(R/W)

Register 38.15. LCD_CAM_LC_DMA_INT_ENA_REG (0x0064)



LCD_CAM_LCD_VSYNC_INT_ENA Write 1 to enable [LCD_CAM_LCD_VSYNC_INT](#). (R/W)

LCD_CAM_LCD_TRANS_DONE_INT_ENA Write 1 to enable [LCD_CAM_LCD_TRANS_DONE_INT](#).
(R/W)

LCD_CAM_CAM_VSYNC_INT_ENA Write 1 to enable [LCD_CAM_CAM_VSYNC_INT](#). (R/W)

LCD_CAM_CAM_HS_INT_ENA Write 1 to enable [LCD_CAM_CAM_HS_INT](#). (R/W)

Register 38.16. LCD_CAM_LC_DMA_INT_RAW_REG (0x0068)

(reserved)																												LCD_CAM_CAM_HS_INT_RAW LCD_CAM_CAM_VSYNC_INT_RAW LCD_CAM_LCD_TRANS_DONE_INT_RAW LCD_CAM_LCD_VSYNC_INT_RAW					
31																												4	3	2	1	0	Reset
0 0																												0	0	0	0		

LCD_CAM_LCD_VSYNC_INT_RAW The raw interrupt status of [LCD_CAM_LCD_VSYNC_INT](#). (RO)

LCD_CAM_LCD_TRANS_DONE_INT_RAW The raw interrupt status of [LCD_CAM_LCD_TRANS_DONE_INT](#). (RO)

LCD_CAM_CAM_VSYNC_INT_RAW The raw interrupt status of [LCD_CAM_CAM_VSYNC_INT](#). (RO)

LCD_CAM_CAM_HS_INT_RAW The raw interrupt status of [LCD_CAM_CAM_HS_INT](#). (RO)

Register 38.17. LCD_CAM_LC_DMA_INT_ST_REG (0x006C)

(reserved)																										LCD_CAM_CAM_HS_INT_ST LCD_CAM_CAM_VSYNC_INT_ST LCD_CAM_LCD_TRANS_DONE_INT_ST LCD_CAM_LCD_VSYNC_INT_ST				
31																								4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

LCD_CAM_LCD_VSYNC_INT_ST The masked interrupt status of [LCD_CAM_LCD_VSYNC_INT](#). (RO)

LCD_CAM_LCD_TRANS_DONE_INT_ST The masked interrupt status of [LCD_CAM_LCD_TRANS_DONE_INT](#). (RO)

LCD_CAM_CAM_VSYNC_INT_ST The masked interrupt status of [LCD_CAM_CAM_VSYNC_INT](#). (RO)

LCD_CAM_CAM_HS_INT_ST The masked interrupt status of [LCD_CAM_CAM_HS_INT](#). (RO)

Register 38.18. LCD_CAM_LC_DMA_INT_CLR_REG (0x0070)

(reserved)																												LCD_CAM_CAM_HS_INT_CLR LCD_CAM_CAM_VSYNC_INT_CLR LCD_CAM_LCD_TRANS_DONE_INT_CLR LCD_CAM_LCD_VSYNC_INT_CLR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
31																									4	3	2	1	0	Reset																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- LCD_CAM_LCD_VSYNC_INT_CLR Write 1 to clear [LCD_CAM_LCD_VSYNC_INT](#). (WO)
- LCD_CAM_LCD_TRANS_DONE_INT_CLR Write 1 to clear [LCD_CAM_LCD_TRANS_DONE_INT](#). (WO)
- LCD_CAM_CAM_VSYNC_INT_CLR Write 1 to clear [LCD_CAM_CAM_VSYNC_INT](#). (WO)
- LCD_CAM_CAM_HS_INT_CLR Write 1 to clear [LCD_CAM_CAM_HS_INT](#). (WO)

Register 38.19. LCD_CAM_LC_REG_DATE_REG (0x00FC)

(reserved)				LCD_CAM_LC_DATE																							
31			28	27																						0	Reset
0	0	0	0	0	0x2502226																						

- LCD_CAM_LC_DATE Version control register. (R/W)